Politecnico di Milano
School of Industrial and Information Engineering
MSc Aeronautical Engineering

Course of DIGITAL CONTROL TECHNOLOGY FOR AERONAUTICS

"Digital Control of a DC Motor–Driven Aircraft Control Surface Positioning System"

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1 Introduction

The project aims to design a digital controller for an aeronautical system. The workflow is developed into two main parts:

1. Design of a discrete-time control algorithm for the chosen system, starting from a set of predefined specifications
2. Evaluation of the proposed control algorithm in a hybrid simulation environment, considering the effect of the various system components

1.1 Problem description

The positioning system of the control surface of an airplane can be modeled as a flexible transmission, in which we have the input tension of the motor (V_a) as the manipulated variable and the position of the output shaft (θ_1) as the controlled variable, the latter being limited in the range [-60°;+60°].

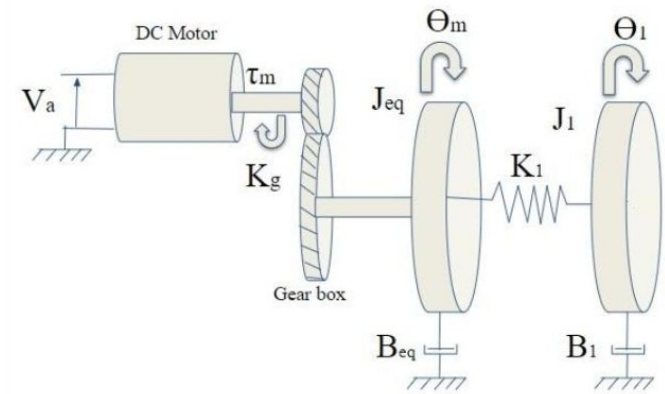


Figure 1: Positioning system of airplane's control surface

The system can be modeled in state space formulation:

$$\dot{x}(t) = \begin{bmatrix} \dot{\theta}_m \\ \ddot{\theta}_m \\ \dot{\theta}_1 \\ \ddot{\theta}_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K1}{J_{eq}} & -\frac{B_{eq}}{J_{eq}} & -\frac{\eta_m \eta_g K_g^2 K_t K_m}{R_a J_{eq}} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{K_1}{J_1} & 0 & -\frac{K_1}{J_1} & -\frac{B_1}{J_1} \end{bmatrix} \begin{bmatrix} \theta_m \\ \dot{\theta}_m \\ \theta_1 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{\eta_m \eta_g K_g K_t}{R_a J_{eq}} \\ 0 \\ 0 \end{bmatrix} V_a \quad (1.1.1)$$

1.2 Data

Symbol	Parameter	Value
R_a	Motor armature resistance	$10\,\Omega$
L_a	Motor armature inductance	$0.18\,\mathrm{mH}$
K_t	Motor current - torque constant	$7.68 \times 10^{-3}\,\mathrm{Nm/A}$
K_m	Motor back-emf constant	$7.68 \times 10^{-3}\,\mathrm{Vs/rad}$
K_g	Total gear ratio	70
η_m	Motor efficiency	0.85
η_g	Gearbox efficiency	0.90
J_{eq}	Equivalent moment of inertia on the motor shaft as seen after the gearbox	$0.0022\,\mathrm{kgm}^2$
B_{eq}	Equivalent viscous damping coefficient on motor shaft	$0.015\,\mathrm{Nms/rad}$
K_1	Flexible coupling stiffness of joint	$1.5\,\mathrm{Nm/rad}$
J_1	Load moment of inertia	$0.025\,\mathrm{kgm}^2$
B_1	Viscous damping coefficient of Load shaft	$0.05\,\mathrm{Nms/rad}$

2 Part 1: continuous time control design

2.1 Controller design

The first part of the project is the design of the controller, therefore the specifications and requirements have to be clearly pointed out. Only then it is possible to proceed with the design phase, clarifying the chosen strategy.

2.2 Requirements for the design of the controller

The controller must regulate the position of the output shaft, satisfying the following conditions:

- tracking of constant angle references with steady-state error lower than 0.3°
- overshoots must be limited to less than 10%
- the speed of the output axis must be limited to +/-45°/s
- the control system must be able to reject constant load torque disturbances applied to the output axis
- the maximum voltage generated by the power amplifier must be limited to the range +/-12V

In addition:

- no requirement is imposed on the settling time and disturbance rejection bandwidth
- the goal is to achieve the fastest possible controlled response while reducing the oscillations on the motor shaft and respecting all other constraints

2.3 Design strategy

2.3.1 Description of the methodology adopted and design criteria

Considering the necessity of limiting the output axis speed and the necessity of rejecting disturbances, the control strategy adopted is a cascade system. In particular, two nested loops are considered: the outer one to regulate the position of the output shaft (θ_1) and the inner one to regulate the speed of the output axis ($\dot{\theta}_1$). The idea is:

- to have an inner loop with a larger rejection bandwidth, so that the efficiency of the system in rejecting disturbances is improved and so the performance
- saturation of the auxiliary variable ($\dot{\theta}_1$) can be handled
- the outer loop controls a slower dynamics with respect to the inner loop, therefore a simpler structure can be chosen, making its realization simpler

The controller design is based on the following criteria:

- the bandwidth of the inner loop must be significantly larger than the bandwidth of the outer loop, i.e.

$$\omega_{c,\text{inner}} \gg \omega_{c,\text{outer}}$$

  typically at least a factor of 2, and preferably up to one order of magnitude;
- the position controller R_{Pos} is implemented as a proportional (P) controller, since the inner velocity loop already provides adequate disturbance rejection;
- the velocity controller R_{Vel} is implemented as a PID regulator to ensure good tracking performance and disturbance attenuation;
- the controller parameters are tuned using a loop-shaping approach in the frequency domain in order to guarantee the required stability margins and performance specifications.

2.3.2 Sampling time

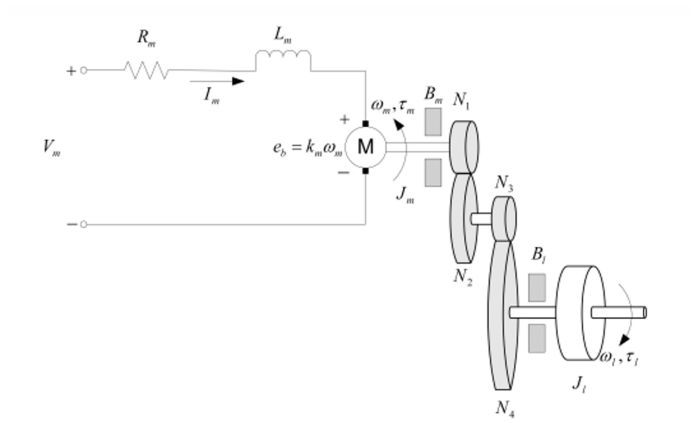


Figure 2: DC motor model

$$\begin{cases} \dot{I}_m(t) = -\frac{R_m}{L_m} I_m(t) - \frac{K_m}{L_m} K_{gear}\omega_l(t) + +\frac{1}{L_m} V_m(t) \\ \dot{\omega}_l = \frac{1}{J_{eq}} \eta K_{gear} I_m(t) - \frac{B_{eq}}{J_{eq}} \omega_l(t) + \frac{1}{J_{eq}} \tau_{ext} \end{cases} \tag{2.3.1}$$

In order to estimate the sampling frequency (thus the sampling time), it's possible to start by considering the closed loop bandwidth, which is approximately equal to the open loop critical frequency. Following an heuristic rule:

$$5\omega_c \leq \omega_s \leq 100\omega_c \tag{2.3.2}$$

The key is the estimation of the open loop critical frequency ω_c. The DC motor is governed by the two equations (2.3.1). Considering the first one:

$\dot{I}_m(t) + \frac{R_m}{L_m} I_m(t) = -\frac{K_m}{L_m} K_{gear}\omega_l(t) + +\frac{1}{L_m} V_m(t)$
Take $\omega_l = 0$
Move to Laplace domain: $sIm(s) + \frac{R_m}{L_m} I_m(s) = \frac{1}{L_m} V_m(s)$
Obtain: $G_{el}(s) = \frac{I_m(s)}{V_m(s)} = \frac{1/R_m}{1+sL_m/R_m}$
From which we obtain:

$$\tau_{el} = \frac{L_m}{R_m} \quad \tau_{el} = 1.8 \times 10^{-5}\,\mathrm{s} \tag{2.3.3}$$

Similarly, for the mechanical part: $\dot{\omega}_l = \frac{1}{J_{eq}} \eta K_{gear} I_m(t) - \frac{B_{eq}}{J_{eq}} \omega_l(t) + \frac{1}{J_{eq}} \tau_{ext}$
Take $I_m = 0$
Move to Laplace domain: $s\omega_l(s) + \frac{B_{eq}}{J_{eq}} \omega_l(s) = \frac{1}{J_{eq}} \tau_{ext}(s)$
Obtain: $G_{mec}(s) = \frac{\omega_l(s)}{\tau_{ext}(s)} = \frac{1/J_{eq}}{1+sB_{eq}/J_{eq}}$
From which we obtain:

$$\tau_{mech} = \frac{J_{eq}}{B_{eq}} \quad \tau_{mech} = 0.147\,\mathrm{s} \tag{2.3.4}$$

The electrical system must be much faster than the mechanical one, as can be seen from the characteristic time constants. Therefore, in order to select the critical frequency of the open loop, the focus is on τ_{mec}. The associated frequency is:

$$\omega_{mec} = \frac{1}{\tau_{mec}} \quad \omega_{mec} \simeq 6.8\,\mathrm{rad/s} \tag{2.3.5}$$

Therefore, according to design procedure employed for the regulator, we obtain a value very close to this: $\omega_c = 6.3\,\mathrm{rad/s}$. Given the heuristic rule (2.3.2), we must have:

$$5 \cdot 6.3 \quad rad/s \leq \omega_s \leq 100 \cdot 6.3 \quad rad/s \tag{2.3.6}$$

So :

$$31.5\,\mathrm{rad/s} \leq \omega_s \leq 630\,\mathrm{rad/s} \tag{2.3.7}$$

Taking the middle value of the interval, we select the sampling frequency and, so, the sampling time: $\omega_s = 315\,\mathrm{rad/s} \Rightarrow T_s = 0.02\,\mathrm{s} = 20\,\mathrm{ms}$.

The open loop critical frequency is, in addition, the approximation of the desired closed loop bandwidth, meaning that also $\omega_{c,inner}$ is defined: $\omega_{c,inner} = 6.3\,\mathrm{rad/s}$. For a cascade control loop, the inner loop must be faster than the outer, meaning that its closed-loop bandwidth must be very much larger, say at least twice the closed-loop bandwidth of the outer loop. Considering to take a larger margin, the choice is: $\omega_{c,inner} = 5\omega_{c,outer} \Rightarrow \omega_{c,outer} = 1.3\,\mathrm{rad/s}$.

2.3.3 Controller design process

Once the sampling time and the closed loop bandwidth of both inner and outer loop are defined, the design phase of the regulator can take place. The first task is to identify a proper PID controller for the inner loop. General transfer function of PID:

$$R_{PID}(s) = K_P + \frac{K_I}{s} + K_D \frac{s}{\tau_f s + 1} \tag{2.3.8}$$

Note that a filtering action is added to the derivative part of the regulator: this is a practical implementation that ensures

- to avoid peaks in the amplification of disturbances, especially high frequency ones
- that the amplification of the steady-state error given by the derivative action does not arrive to the actuator, especially in the case it is very sensible and cannot withstand intense inputs.

It is possible to see that the PID is equivalent to a regulator with one pole in the origin, one zero and a zero-pole pair. The loop-shaping design process, consists in finding proper values of those poles and zeros so that requirements are satisfied: at first, we set the filtering action on the derivative part, so all the poles are well-defined (the other one is in the origin).

$$R_{PID}(s) = \mu \frac{(1+\tau_1 s)(1+\tau_2 s)}{(1+\tau_f s)} \tag{2.3.9}$$

The filtering action has to act on high frequencies, above the bandwidth of the inner loop, to reject all the frequencies that would cause an undesired amplification of steady-state error. Knowing that $\omega_{c,inner} = 6.3\,\text{rad/s}$, it is possible, for example, to set ω_f as 5-10 $\omega_{c,inner}$. Take $\omega_f = 5\omega_{c,inner}$, therefore:

$$\omega_f = 5 \cdot 6.3\,\text{rad/s} = 31.5\,\text{rad/s} \Rightarrow \tau_f = 1/\omega_f = 0.0317\,\text{s} \tag{2.3.10}$$

Being more conservative, it is possible to choose $\tau_f = 0.02\,\text{s}$.

The remaining parameters that have to be tuned by loop shaping are the two zeroes and the gain. In order to satisfy the requirement on the phase margin, the choice is:

- $\tau_1 = 0.15\,\text{s}$
- $\tau_2 = 0.3\,\text{s}$
- $\mu = 20$

Finally:

$$R_{PID}(s) = 20 \frac{(1+0.15s)(1+0.3s)}{(1+0.02s)} \tag{2.3.11}$$

Once the inner regulator is defined, the next step is the design of the outer regulator, which is a proportional one. Given that the bandwith of the inner loop is much larger than the one of the outer loop, it is possible to assume that the behavior of the inner loop is seen as a proportional gain: therefore the only operation that must be done to set the outer loop regulator is to properly scale this proportional gain according to $\omega_{c,outer}$.
$G_{\theta_1}(s) = \mu_{inner}/s$
$L_{\theta_1}(s) = R_P(s)G_{\theta_1}(s) = \frac{K_{PO}\mu_{inner}}{s}$
Imposing $|L_{\theta_1}(j\omega_{c,outer})| = 1$, we find: $K_{PO} = \frac{\omega_{c,outer}}{\mu_{inner}}$, with μ_{inner} being the gain of F_{inner}.
Therefore:

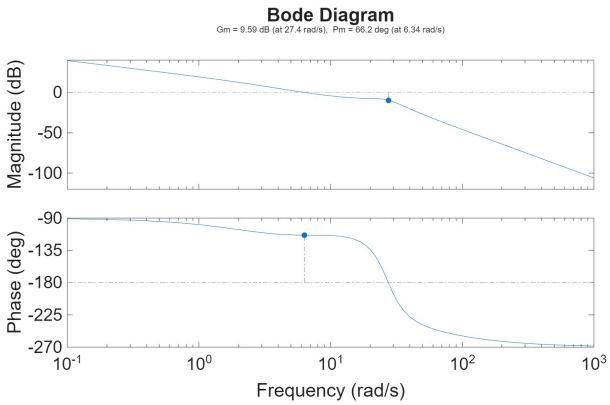
$$R_P(s) = 1.3 \tag{2.3.12}$$

2.3.4 Description of controllers and closed loop transfer functions

Given the designed PID and P controllers, it is possible to define and analyze the closed loop transfer functions of both inner and outer loops.

2.3.5 Stability margins

From the closed-loop transfer functions, it is possible to analyze the stability margins of the system and, thus, the satisfaction of the requirements.



(a) L_{inner} margins

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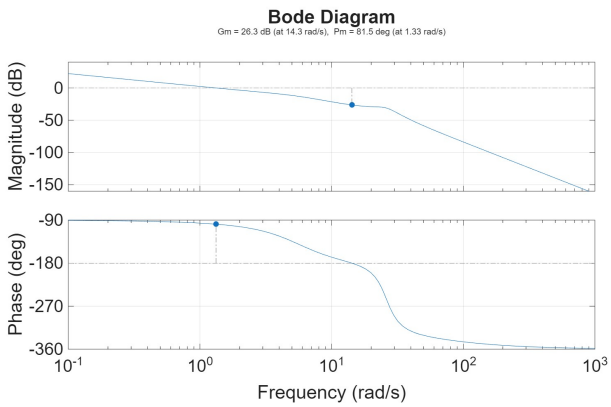


(b) L_{outer} margins

Figure 3: Closed loop transfer functions' stability margins

The phase margin in both loops respects the limits imposed by project data: for the inner loop $PM_{inner} = 66.2°$, while for the outer loop $PM_{outer} = 81.5°$, meaning that in both cases the phase margin is higher than 45°. The following plot shows the step response of the outer closed-loop system, assuming the inner loop is represented by its static gain: it is possible to observe that there is no overshoot.

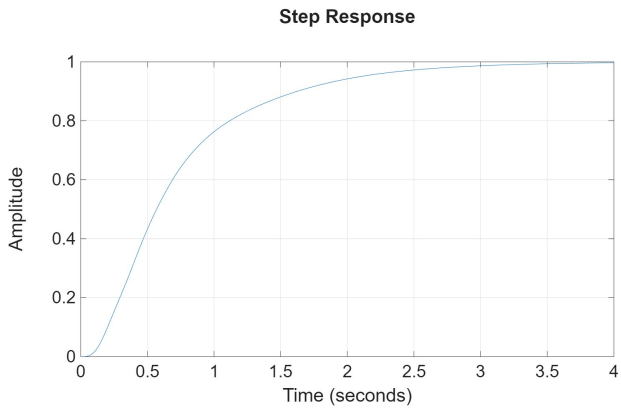


Figure 4: Step response of the outer closed-loop system

3 Part 2: digital control system implementation

3.1 Signal acquisition system parameters

To implement the digital controller, the following hardware is available:

- ADC converters with a voltage span of [0, 3.3V], selectable resolution between 8 and 12 bits, and conversion time 1 μs.
- A couple, DAC - Power Amplifier, to drive the DC motor, taking as input a digital (integer) value in the range [0, 1023] and producing a voltage in the range [-12V , 12V], with a negligible response time.
- Both axes possess optical encoders measuring the shaft's position with a resolution of 2048 counts per revolution, and tachometers for speed measurement with a sensitivity of 10mV/(°/s), span [-60°/s, 60°/s], and 0V reading for zero speed condition.

Once the control law is designed in continuous time, a hybrid simulation framework is implemented in order to evaluate the performance of the digital control system, taking into account the effects of signal acquisition, digitization, processing, and actuation.

3.2 Controller discretization and sampling effects

The continuous-time controllers are discretized using the Tustin (bilinear) transformation, while the plant is discretized using a Zero-Order Hold (ZOH) assumption, with sampling time:

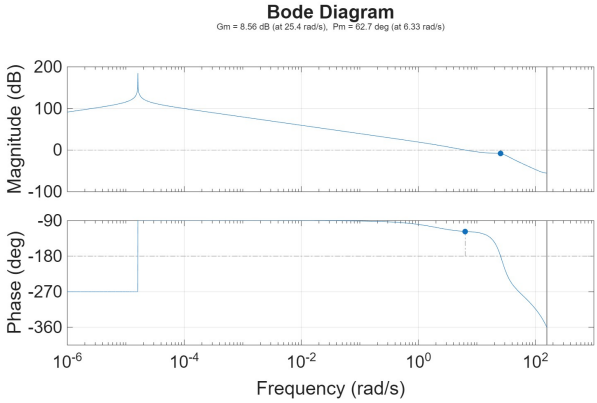
$$T_s = 0.02\,\text{s}. \tag{3.2.1}$$

Both inner (velocity) and outer (position) loops are analyzed in the discrete domain by evaluating open-loop frequency responses and closed-loop step responses.

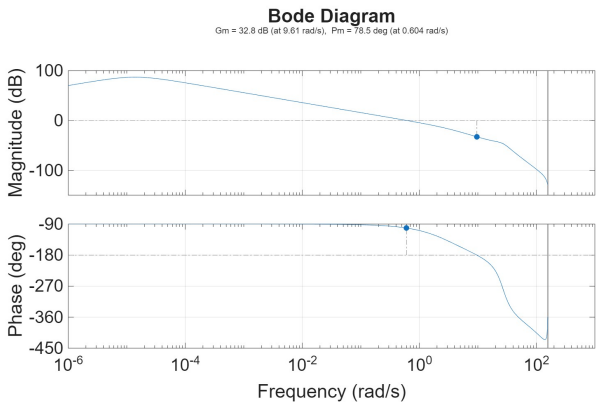
The phase loss introduced by sampling is estimated as:

$$\Delta\phi \approx \frac{1}{2}\omega_c T_s, \tag{3.2.2}$$

with ω_c being the crossover frequency. This effect is verified for both loops, confirming the degradation of phase margin due to discretization.



(a) L_{inner} margins



(b) L_{outer} margins

Figure 5: Closed loop transfer functions' stability margins after discretization

3.3 Anti-aliasing filter design

To prevent aliasing, a second-order Butterworth low-pass filter is designed. The sampling frequency and Nyquist frequency are:

$$\omega_s = \frac{2\pi}{T_s} \approx 314.16\,\text{rad/s}, \qquad \omega_N = \frac{\omega_s}{2} \approx 157.08\,\text{rad/s}. \tag{3.3.1}$$

The anti-aliasing filter cutoff is placed at one-third of the Nyquist frequency:

$$\omega_{AA} = \frac{\omega_N}{3} \approx 52.36\,\text{rad/s}, \qquad \tau_{AA} = \frac{1}{\omega_{AA}} \approx 0.019\,\text{s}. \tag{3.3.2}$$

The resulting second-order transfer function is:

$$G_{AA}(s) = \frac{1}{(\tau_{AA}\, s + 1)^2}. \tag{3.3.3}$$

This filter is included in both feedback paths and its effect on the phase and gain margins of the open-loop transfer functions is verified via Bode analysis.

3.4 Signal conditioning and ADC modeling

The signal conditioning stage can be designed starting from the sensor specifications. Note that the LSB of the acquisition system has to guarantee that the minimum voltage that can be converted is lower than the sensitivity voltage of the instrument, in order to

not have a loss in terms of precision of the digitalization process. Therefore, for the choice of the resolution of the ADC, two approaches are followed:

- inner loop: the sensitivity of the tachometer is $10\,\mathrm{mV}/(^\circ/\mathrm{s})$. With 8 bit- architecture:

$$LSB_8 = V_{fs}/(2^8 - 1) = 12.9mV > 10mV \tag{3.4.1}$$

  So an architecture with at least 9 bit is required, such that $LSB_9 = 6.46mV < 10mV$;

- outer loop: considering that the optical encoder works with 2048 counts/rad, an 11-bit architecture for the ADC is adopted.

3.4.1 Inner loop (velocity measurement)

The inner loop measures angular velocity through a tachometer with sensitivity $K_{\text{tach}} = 10\,\mathrm{mV}/(^\circ/\mathrm{s})$, corresponding to:

$$K_{\text{transducer,vel}} = \frac{10 \times 10^{-3}\,\mathrm{V}}{\pi/180\,\mathrm{rad/s}} \approx 0.5730\,\frac{\mathrm{V}}{\mathrm{rad/s}}. \tag{3.4.2}$$

The tachometer output spans $[-600, +600]\,\mathrm{mV}$ for the full velocity range of $\pm 60\,^\circ/\mathrm{s}$. An op-amp conditioning stage maps this range to the ADC input range $[0, 3.3]\,\mathrm{V}$, applying:

- Amplification gain: $K_{\text{op-amp}} = 2.75\,\mathrm{V/V}$,
- Bias (offset): $V_{\text{bias}} = 1.65\,\mathrm{V}$.

The conditioned signal is then filtered by $G_{AA}(s)$ and converted by a 9-bit ADC ($V_{\text{fs}} = 3.3\,\mathrm{V}$):

$$\mathrm{LSB}_9 = \frac{V_{\text{fs}}}{2^9 - 1} = \frac{3.3}{511} V/bit \tag{3.4.3}$$

The quantized output is then normalized back to physical units in the sensor driver block:

$$V_{\text{out}} = \frac{V_{\text{fs}}}{2^9 - 1} \cdot n - 1.65\,\mathrm{V}, \qquad \dot{\theta}_1 = V_{\text{out}} \cdot \frac{60\,^\circ/\mathrm{s}}{1.65\,\mathrm{V}}, \tag{3.4.4}$$

where n is the ADC integer output.

3.4.2 Outer loop (position measurement)

The outer loop measures angular position via an optical encoder with a resolution of 2048 counts per revolution. The encoder count is mapped to the ADC range $[0, 3.3]\,\mathrm{V}$ through:

- Encoder gain: $\dfrac{2048}{2\pi/3}$ counts/rad (for a $\pm 60^\circ$ range),
- Offset: $+1024$ counts (to center the range),
- Scaling to voltage: $\dfrac{3.3}{2048}\,\mathrm{V/count}$.

After anti-aliasing filtering, an 11-bit ADC is used:

$$\mathrm{LSB}_{11} = \frac{V_{\text{fs}}}{2^{11} - 1} = \frac{3.3}{2047} V/bit \tag{3.4.5}$$

The sensor driver reconstructs the physical angle:

$$V_{\text{out}} = \frac{V_{\text{fs}}}{2^{11} - 1} \cdot n - 1.65\,\mathrm{V}, \qquad \theta_1 = V_{\text{out}} \cdot \frac{\pi/3\,\mathrm{rad}}{1.65\,\mathrm{V}}. \tag{3.4.6}$$

3.5 DAC and actuation modeling

The DAC–Power Amplifier chain converts the digital control output to the motor armature voltage $V_a \in [-12, +12]\,\mathrm{V}$. The quantizer resolution is set to:

$$\mathrm{LSB}_{\mathrm{DAC}} = \frac{12}{2^9 - 1} \approx 23.48\,\mathrm{mV/bit}. \tag{3.5.1}$$

In the Simulink scheme, the output of each controller passes through: a quantizer with step $\mathrm{LSB}_{\mathrm{DAC}}$, a Zero-Order Hold with period T_s, and a computational delay of $T_s/2$ to model the processing latency.

3.6 Simulink hybrid simulation scheme

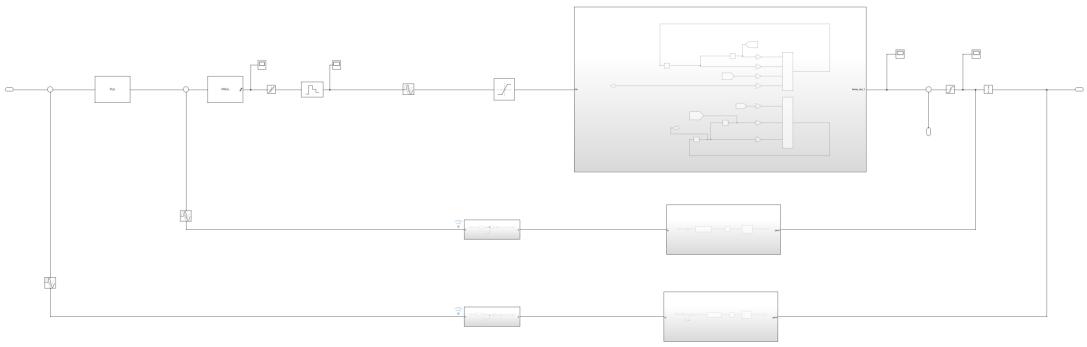


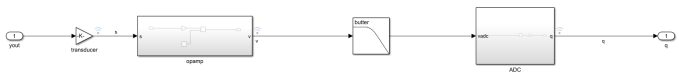
Figure 6: Simulink scheme

The complete hybrid simulation was implemented in MATLAB/Simulink R2024b. The feedback path of each control loop includes the following blocks in series:

1. **ADC + Conditioning block**: sensor transducer gain → op-amp amplification and bias → second-order anti-aliasing filter $G_{AA}(s)$ → quantizer → normalization to ADC counts.

2. **Sensor driver block**: conversion from ADC counts back to physical units (velocity in rad/s or position in rad), with sign and offset correction.

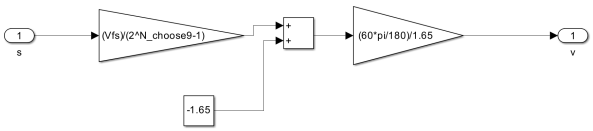
3. **ADC delay block**: pure delay modeling the conversion time ($T_{\mathrm{delay,vel}} = 9bit \cdot 1\mu s/bit = 9\,\mu\mathrm{s}$, $T_{\mathrm{delay,pos}} = 11bit \cdot 1\mu s/bit = 11\,\mu\mathrm{s}$).



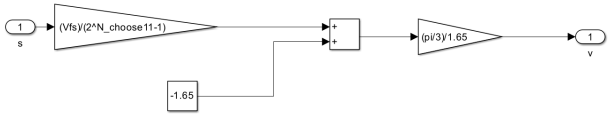
(a) ADC velocity

(b) ADC position

Figure 7: ADC blocks



(a) Sensor driver velocity



(b) Sensor driver position

Figure 8: Sensor driver blocks

On the actuation side, after each controller:

1. **DAC quantizer**: step $= \mathrm{LSB}_{\mathrm{DAC}}$.
2. **Zero-Order Hold**: sample period T_s.
3. **Computational delay**: $T_s/2$, modeling the processor execution latency.

3.7 Performance evaluation

The performance of the complete hybrid digital control system is evaluated through a series of simulation tests designed to verify compliance with all the design requirements.

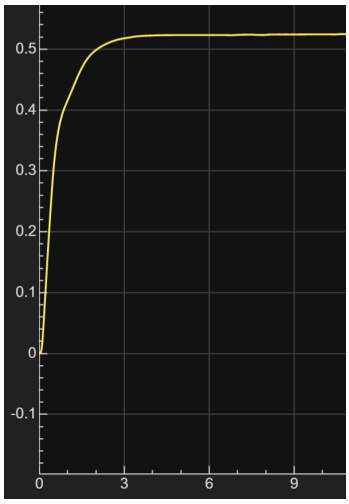
3.7.1 Step response and tracking performance

A step reference of $\theta_1^{\mathrm{ref}} = 30\,^\circ$ was applied to the outer position loop in order to assess tracking performance under nominal conditions. The following quantities were extracted from the simulation:

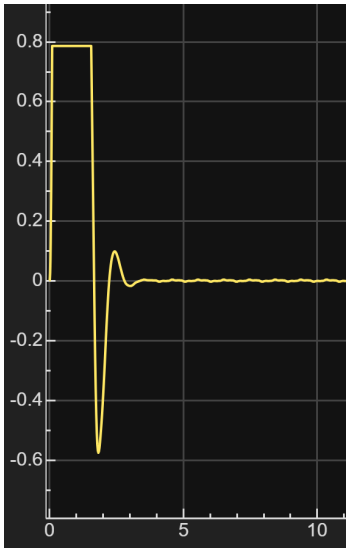
- **Steady-state error**: the residual position error after the transient has settled to $0°$, verified to be lower than $0.3\,^\circ$ as required.
- **Overshoot**: the percentage overshoot on $\theta_1(t)$, verified to be below $10\,\%$.
- **Output axis speed**: the time history of $\dot{\theta}_1(t)$, verified to remain within $\pm 45\,^\circ/\mathrm{s}$ at all times.

3.7.2 Velocity saturation and anti-windup

To verify the correct operation of the velocity saturation block, a large position step ($\theta_1^{\mathrm{ref}} = 60\,^\circ$, i.e. the maximum allowable angle) is applied. Under this condition the inner-loop reference $\dot{\theta}_1^{\mathrm{ref}}$ generated by the outer P controller exceeds the $\pm 45\,^\circ/\mathrm{s}$ limit, activating the saturation. The anti-windup mechanism prevents integrator wind-up in the PID velocity controller, ensuring a smooth recovery once the saturation is reached.



(a) Step response of the outer closed-loop system. Reference: $\theta_1^{\text{ref}} = 30\,^\circ$.



(b) Velocity saturation test: step reference of 60°.

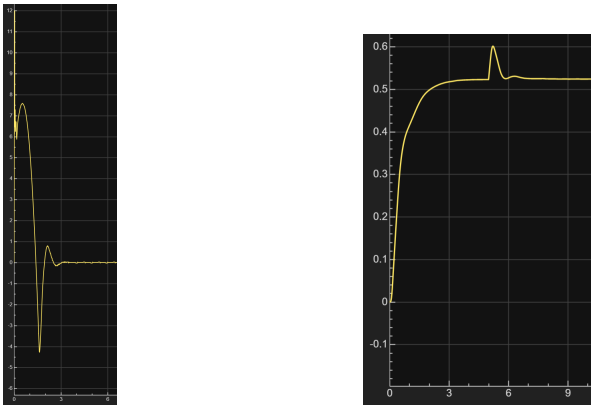
Figure 9: Step response of outer loop and velocity saturation

3.7.3 Actuator voltage constraint

The control voltage $V_a(t)$ applied to the DC motor must remain within $\pm 12\,\text{V}$ at all times, including during transients. The DAC-Power amplifier chain is modeled with a saturation block at $\pm 12\,\text{V}$ (corresponding to integer values [0, 1023] at the DAC input). The following plot shows the evolution of $V_a(t)$ for the large-step test of Section 3.7.2, confirming that the voltage constraint is never violated.

3.7.4 Disturbance rejection

A constant disturbance of $\omega_d = 0.5\,\text{rad/s}$ is applied to the output shaft once the system has reached steady state at $\theta_1^{\text{ref}} = 30°$. The disturbance is expressed as an angular velocity since the plant output is itself an angular velocity; the underlying physical quantity is a constant load torque, which can be equivalently described as $\tau_d = \omega_d J_1$, where J_1 is the inertia. The integral action embedded in the inner PID velocity controller drives the steady-state position error to zero, ensuring full rejection of constant load torques as required.



(a) (b)

Figure 10: (a) Control voltage $V_a(t)$ during the large-step test. The signal remains within the $\pm 12\,\mathrm{V}$ hardware limit; (b) Disturbance rejection test. A constant angular velocity disturbance $\omega_d = 0.5\,\mathrm{rad/s}$ is applied to the output shaft at $t = 5\,\mathrm{s}$.

4 Observations and conclusions

The digital control system is, in conclusion, able to:

- track constant angle references with a steady state error that goes to zero, thus below $0.3°$
- limit the overshoot under a value of 10%
- limit the speed at output axis to $\pm 45°/s$ thanks to a saturation block inserted in the model after the system has produced an angular velocity output
- successfully reject constant load torque disturbances
- limit the maximum input voltage to the system to $\pm 12V$ using a saturation block before the voltage input enters the system dynamics

This means that all project requirements are met, with a settling time that seems to be acceptable in real terms (almost 3 seconds to deflect an aircraft surface by 30°). In addition, the oscillations that can appear are definitely acceptable if compared with reference values.

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